COMP4136 Mini-Project

Requirements:

1. You should form into groups of **one to three** students to carry out this project.
2. Timeline
   a) **By 5:00pm, 6 Nov 2026 (Friday):** Please submit your group member(s), student ID(s), and the selected topic in the Google sheet (https://docs.google.com/spreadsheets/d/1sIiF9HRM8axMXYOHcu95UTKYZ-BUAKF4ZInmqrU9BVU/edit?usp=sharing. The group order is your group number. (No need to send the group information to TA by email)
   b) **By 5:00pm, 20 Nov 2026 (Friday):** You should submit the **final project report and a Powerpoint file** via Moodle before this deadline. Your Powerpoint should show the background, methods and main results of your project in a format that is easy to follow and understand.
   c) The project presentation will be conducted **onsite** at AST916&RRS638 (tentatively at Nov 23&26). The order with Group numbers will be released later. Your **10-min presentation** should show the main results of your mini-project in a format that is easy to follow and understand. During the presentation, each group member should clearly present his/her own work and contribution in this project and the entire presentation should be organized in a logic way.
3. You are required to submit two files for your project: (1) the final project report, in .pdf format, and (2) one Powerpoint file in .pdf format. Please compress the two files as one final submission file, and name the final submission file in the format of "Group_[group number].zip". For example, if the group number is 8, the file name should be "Group_8.zip"
4. **Any form of plagiarism is strictly prohibited. Students who plagiarized and whom were plagiarized will be given zero mark and reported to the departmental exam committee for further penalty.**

Topic Selection

For topic selection, you may:

- Choose a topic from the list below, where we provide **12** topics;
- or Propose your own topic, but it must be related to NLP/LLM.
  - Statistical Language Models and Neural Language Models
  - Traditional Word Embeddings and Contextual Embedding
  - Text Summarization
  - Sentiment Analysis
  - Machine Translation
  - Question answering
  - Textual Visualization
  - Adapt LLMs to a new Language
  - Vertical domain LLMs

- Evaluation of LLMs
- Build Dataset for pre-training LLMs
- A survey for a specific topic

Overall Picture

This project motivates students to dive into the rapidly evolving field of Natural Language Processing (NLP) and Large Language Model (LLM). It might involve:

(1) For a selected topic of interest, conduct a comprehensive survey on its background;
(2) Understand the challenges in this filed and identify the primary properties of existing technologies;
(3) Clearly define problem(s) related to the topic;
(4) Prepare the data, implement or propose innovative approaches to push the field forward.

We provide *two examples* for reference in Appendix A.

Project Template:

The project report should include, but is not limited to, the following sections:

1) **Abstract**: A concise summary of the project, including the problem statement, methodology, main findings, and conclusions.
2) **Background Survey**: Conduct a literature review to understand the development of your chosen topic, key milestones achieved, and the current state-of-the-art. Analyze existing methodologies to identify their limitations.
3) **Methodology Implementation**: Implement one or more existing methods on the selected topic, or propose a new method addressing identified limitations. Your approach should be well-justified.
4) **Experiments**: Describe the data used, including its source, characteristics, and any pre-processing steps. Conduct experiments to evaluate the effectiveness of the chosen/developed method(s). Show experimental results and Discuss the significance of your findings.
5) **Case Study/Application**: Present a case study or a practical example of your project.
6) **Author Contributions**: Contribution of each group member in this project, in terms of the specific task carried out.
7) **References**: Please refer to the IEEE website (https://pitt.libguides.com/citationhelp/ieee) for reference citation style (book, chapter in book, journal article, conference paper, etc).

Scoring Rubrics:

The mini-project consists of two parts: Project Presentation and Project Report.

- Project Presentation:
  - Content quality (50%): Well-presented slides are highly valued.
  - Oral presentation (50%): Clear and enthusiastic speaking is encouraged.
- Project Report:
  - Technical excitement (20%): It is encouraged to do something that is either interesting or useful!
  - Technical soundness (50%): A) Discuss the motivation on why you work on this project and your algorithm or approach. Even if you are reproducing a published paper, you should have your own motivation. B) Cite existing related work. C) Present your algorithms or systems for your project. Provide key information for reviewers to judge whether it is technically correct. D) Provide a reasonable evaluation protocol, detailed enough to contextualize your results. E) Report quantitative results and include qualitative evaluation. Analyze and understand your system by inspecting key outputs and intermediate results. Discuss how it works, when it succeeds and fails, and try to interpret why it works and why not.
  - Clarity in writing (20%): The report is written in a precise and concise manner so the report can be easily understood.
  - Individual contribution (10%): This is based on individual contribution, probably on a subjective basis.

Appendix A

1. Vertical Domain LLMs

The domain of Vertical large language models (LLMs) encompasses a wide range of subjects including Medicine, Law, Finance, Software Engineering, Science, Geometry, Math, Optimization, and Education, among others. The challenge in this field lies in effectively injecting new knowledge into an existing LLM, which requires a comprehensive pipeline that integrates continuous learning and updating mechanisms. This process must ensure that the model can assimilate and apply the newly introduced information accurately and efficiently, while maintaining its existing knowledge base and performance. It involves sophisticated algorithms and techniques to ensure that the LLM remains relevant and enhances its capabilities without compromising the integrity of its existing structure and knowledge.

To implement your own model, you can consider some techniques, such as:

- **Supervised-tuning**: Fine-tune the LLM with a smaller, task-specific dataset, which includes labeled examples to guide the model towards desired outputs. Data collection is a critical step that precedes both pre-training and fine-tuning, where high-quality corpora, either structured or unstructured, are assembled to serve as the foundation for model learning. Self-instruction is a technique where the model generates prompts and responses, creating a dataset that can be used for further training to improve the model's ability to follow instructions and generate relevant outputs.
- **Retrieval-Augmented Generation (RAG)**: Incorporate external knowledge to enhance the model's factual accuracy and generative capabilities.

2. Evaluation of LLMs

Investigate the large language models (such as ChatGPT, Qwen, LlaMA, Claud, GPT-3.5, DeepSeek, etc) to assess their capabilities, limitations, and potential risks. You are also encouraged to develop your own evaluation methods and datasets, assessing the performance of LLMs in a field you're familiar with. Your investigation may include, but is not limited to, the following areas:

- **Math reasoning**: evaluate LLMs' ability to solve mathematical problems and perform complex reasoning tasks.
- **Hallucination**: LLMs may generate inaccurate or false information, known as hallucinations. Assess the factual accuracy and reliability of generated content, and develop methods to detect and mitigate such errors.
- **Long-context evaluation**: Test the model's ability to understand and retain long-range dependencies within extended texts. This is crucial for applications involving in-depth conversations, document understanding, or narrative coherence.
- **Multilingual Capability**: Assess the model's proficiency in processing and integrating content across multiple languages, including understanding, translation, and cross-lingual reasoning.

- **Safety and privacy risks**: Evaluate the model's vulnerability to generating harmful content or leaking sensitive information. The focus should be on developing safer systems that prioritize user protection and data privacy.

Each of these dimensions contributes to ensuring that LLMs can operate effectively, responsibly, and safely across a broad range of real-world applications.

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